CAMBRIDGE IGCSE CHEMISTRY 0620

Topic 7: Acids, bases and salts

Premium Answer Booklet

23 questions	2020-2025 Paper 4	Premium readable layout

Rebuilt for easier reading with structured tables, clear answer sections and highlighted workings. Use beside the matching topical question bank.

Contents

Question 01

16 marks

2020 May/June Paper 41 Question 4

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 3: Stoichiometry, Topic 12: Experimental techniques and chemical analysis
Answer source label: 0620_s20_ms_41.pdf

4(a)	• (damp) litmus • (turns) blue
4(b)(i)	• proton acceptor
4(b)(ii)	• Above pH 7 up to 11
4(b)(iii)	• blue precipitate • precipitate dissolves • deep blue solution remains
4(c)(i)	• neutralisation
4(c)(ii)	• Na2SO4 • 2H2O
4(d)(i)	• methyl orange
4(d)(ii)	• mol of NaOH • (•) • 25.0 • 0.0400 • 0.001 00 • = • x • = • mol • mol of H2SO4 • (•) • 0.001 • 0.0005 00 • = • = • = • 1000 • 0.0005 • 0.025 • 20.0 • 20.0 • x • = • x • = • (mol / dm3) • error carried forward may be allowed
4(d)(iii)	• use of 40 g/mol • 40 x 0.04 = 1.6 (g/dm3)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 02

13 marks

2020 Oct/Nov Paper 41 Question 3

Main topic: Topic 7: Acids, bases and salts
Answer source label: 0620_w20_ms_41.pdf

3(a)(i)	• 2 -> 2 + 2
3(b)	• 75(%)
3(c)	• test: (damp red) litmus paper (1) • result: (litmus goes) blue (1)
3(d)(i)	• diffusion
3(d)(ii)	• particles move from an area of high to low concentration • particles move randomly
3(d)(iii)	• CO2 molecules are heavier (than NH3)
3(e)(i)	• lower temperature: (rate of reaction) slower (1) • higher pressure: expensive/specialist equipment
3(e)(ii)	• catalyst
3(f)(i)	• proton acceptor
3(f)(ii)	• any value greater than 7 up to 12

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
• escape.

END OF ANSWER

Question 03

26 marks

2020 Oct/Nov Paper 42 Question 2

Main topic: Topic 7: Acids, bases and salts
Answer source label: 0620_w20_ms_42.pdf

2(a)	• HNO_3
2(b)(i)	• to make sure all the (sulfuric) acid reacts
2(b)(ii)	• no (more) fizzing (1) • ($FeCO_3$) stops dissolving or a solid remains / is visible (in the mixture) (1)
2(b)(iii)	• rinse the residue (with distilled water)
2(b)	• (iv • a solution that can dissolve no more solute (1) • at the specified temperature (1)
2(b)(v)	• iron(II) oxide / iron(II) hydroxide
2(c)	• mass of $FeSO_4$ = 152 (1) • mass of H_2O = 278 - 152 = 126 (1) • mol of H_2O = 126 / 18 and x = 7 (1)
2(d)(i)	• precipitation
2(d)(ii)	• cream precipitate
2(d)(iii)	• $Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$ • AgBr (as only product) (1) • Ag^+ and Br^- (as reactants)(1) • state symbols(1)
2(e)	• mol of NaCl = 2.34 / 58.5 = 0.04(00) • mol of Cl_2 = /2 = 0.04(00)/2 = 0.02(00) • 0.02(00) x 24000 = 480 (cm^3)
2(f)(i)	• ions (1) • (ions) are fixed (in a lattice) (1) • ions are mobile (1)
2(f)(ii)	• chlorine
2(f)(iii)	• oxidation (1) • electrons are lost (1)
2(f)(iv)	• dissolve it (in water)

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 04

17 marks

2021 Feb/March Paper 42 Question 4

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 3: Stoichiometry, Topic 12: Experimental techniques and chemical analysis
Answer source label: 0620_m21_ms_42.pdf

4(a)	• $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$ • $ZnCl_2$ or H_2 as a product (1) • correct equation (1) • states (1)
4(b)	• to make sure all the (hydrochloric) acid reacts
4(c)	• filtration
4(d)	• a solution that can dissolve no more solute (1) • at a given temperature (1)
4(e)	• solubility (of $ZnCl_2$ / solids) decreases (as temperature decreases)
4(f)	• zinc oxide • zinc carbonate
4(g)	• Ca will also react with water
4(h)(i)	• neutralisation
4(h)(ii)	• 0.100 x 25 / 1000 = 0.0025(0) (1) • 0.0025 x 2 = 0.005(00) (1) • 0.005 x 1000 / 20 = 0.25(0) (1) • Mr = 40 (1) • 0.25 x 40 = 10.(0) (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 05

6 marks

2021 May/June Paper 41 Question 1

Main topic: Topic 7: Acids, bases and salts
Answer source label: 0620_s21_ms_41.pdf

1(a)	• Haber (process)
1(b)	• fractional distillation
1(c)	• electrolysis
1(d)	• filtration
1(e)	• hydrolysis
1(f)	• chromatography

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 06

15 marks

2021 May/June Paper 41 Question 5

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_s21_ms_41.pdf

5(a)	• (add) water (to both salts) (1) • dissolve both salts / make solutions (1) • filter (lead(II) iodide)(1) • wash (residue of lead(II) iodide) with water AND dry e.g. with filter paper / description of washing and drying (1) • $Pb(NO_3)_2 + 2\ NaI \rightarrow 2NaNO_3 + PbI_2$ • OR $Pb^{2+} + 2I^- \rightarrow PbI_2$ (1)
5(b)(i)	• glowing splint (1) • relights / rekindles (1)
5(b)(ii)	• $2ZnO(s)$ and $4NO_2(g)$ (1) • $12H_2O(g)$ (1)
5(c)(i)	• heat again and weigh again / repeat steps 2 and 3 (1) • until mass is constant (1)
5(c)(ii)	• 0.005 (1) • 0.9 (1) • (0.9 / 18 =) 0.05 (1) • (0.05 / 0.005 =) 10 (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 07 — 15 marks

2021 May/June Paper 43 Question 5

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_s21_ms_43.pdf

5(a)	• add zinc carbonate to sulfuric acid until • it stops dissolving • OR • no more effervescence (1) • filter (zinc carbonate) (1) • evaporation of filtrate to form dry crystals (1) • $ZnCO3 + H2SO4 -> ZnSO4 + CO2 + H2O$ • CO2 product in equation (1) • rest of equation correct (1)
5(b)(i)	• acidified aqueous potassium manganate(VII) (1) • purple to colourless (1)
5(b)(ii)	• 2 FeSO4.7H2O (1) • 14 H2O (1)
5(c)(i)	• heat + weigh + repeat (1) • until mass is constant (1)
5(c)(ii)	• 0.02 (1) • 0.72 (1) • / 18 = 0.72/18 = 0.04 (1) • / = 0.04 / 0.02 = 2 (1) • Question Answer

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 08 — 10 marks

2021 Oct/Nov Paper 42 Question 2

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_w21_ms_42.pdf

2(a)(i)	• strong
2(a)(ii)	• 2H+ + SO42- • H+ (1) correct equation (1)
2(a)(iii)	• pink / red
2(b)(i)	• ZnCO3(s) + 2HNO3(aq) -> • Zn(NO3)2(aq) + H2O(l) + CO2(g) • reactant states (1) product states (1)
2(b)(ii)	• 125 • 2.5 / 125 = 0.02(00) • 0.02(00) x 2 = 0.04(00) • 0.04(00) x 1000 / 20 = 2(.00)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 09 — 11 marks

2021 Oct/Nov Paper 43 Question 5

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 12: Experimental techniques and chemical analysis
Answer source label: 0620_w21_ms_43.pdf

5(a)(i)	• too many colour changes
5(a)(ii)	• Any acid-base indicator, e.g. methyl orange or phenolphthalein
5(b)	• repeat without indicator using same volumes OR remove indicator by adding charcoal or carbon and filtering (1) • evaporate / heat / warm/ • boil/leave in hot place (1) • until most of the water is gone / some water left / • saturation(point) / • crystallisation (point) / • evaporate some of the water (1) • cool / • leave to crystallise(1) • description of drying (1)
5(c)(i)	• yellow flame
5(c)(ii)	• solid dissolves / disappears(1) • blue solution(1)
5(c)(iii)	• white precipitate

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 10 | 11 marks

2022 May/June Paper 41 Question 6

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 9: Metals
Answer source label: 0620_s22_ms_41.pdf

6(a)(i)	• zinc blende
6(a)(ii)	• heat(zinc sulfide) strongly in air / roast in air
6(a)(iii)	• carbon or carbon monoxide
6(a)(iv)	• the temperature in the furnace is above or higher than the boiling point of zinc • OR • the boiling point of zinc is below or less than the temperature of the furnace
6(a)(v)	• condensation / condensing / condense
6(b)(i)	• no bubbles • or • no fizzing • or • no effervescence
6(b)(ii)	• zinc / zinc oxide / zinc hydroxide
6(b)(iii)	• (powder has) larger surface area OR lumps have smaller surface area (1) • (powder has) more collisions per unit time / more collision frequency • OR • lumps have fewer collisions per unit time / less collision frequency (1)
6(b)(iv)	• crystals (form on glass rod or microscope slide)
6(b)(v)	• ZnSO4

Easy explanation / reminder
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 11 — 23 marks

2022 Oct/Nov Paper 41 Question 2

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 3: Stoichiometry, Topic 2: Atoms, elements and compounds
Answer source label: 0620_w22_ms_41.pdf

2(a)	• metallic (1) • lattice of potassium ions (1) • sea of electrons (1) • attraction between potassium ions and electrons (1)
2(b)(i)	• any two (one from each bullet point) • physical constants: high boiling point / melting point • conductivity: conduct electricity when aqueous / conduct electricity when molten • solubility: soluble in water
2(b)(ii)	• eight dots in third shell of both K (1) • six crosses and two dots in third shell of S (1) • '+' charge on each K on correct answer line and '2-' charge on S ion on correct answer line (1)
2(c)(i)	• lilac
2(c)(ii)	• OH^-
2(c)(iii)	• blue
2(c)(iv)	• mol of K = 2.34 / 39 = 0.06(00) (1) • mol of H_2 = 0.06 / 2 = 0.03(00) (1) • volume of H_2 = 0.03 -> 24 000 = 720 cm^3 (1)
2(d)(i)	• hydrochloric (acid)
2(d)(ii)	• neutralisation
2(d)(iii)	• titration
2(e)(i)	• white
2(e)(ii)	• silver chloride
2(e)(iii)	• $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$ • AgCl (as only product) (1) • Ag^+ and Cl^- (as only reactants) (1) • state symbols (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 12 | 13 marks

2023 May/June Paper 41 Question 6

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 10: Chemistry of the environment
Answer source label: 0620_s23_ms_41.pdf

6(a)(i)	• $4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$
6(a)(ii)	• iron(III) oxide
6(b)(i)	• yield of SO_3 is less
6(b)(ii)	• yield of SO_3 is less • OR • rate is less
6(c)	• $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$ • $(NH_4)_2SO_4$ on the right (1) • equation correct(1)
6(d)(i)	• lead(II) nitrate
6(d)(ii)	• $Pb^{2+}(aq) + SO_4^{2-}(aq) \rightarrow PbSO_4(s)$ • $PbSO_4$ on the right(1) • only Pb^{2+} and SO_4^{2-} on the left(1) • (aq) + (aq) -> (s)(1)
6(d)(iii)	• filter(1) • wash (the residue or lead sulfate) with distilled or deionised water(1) • description of drying(1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
• escape.

END OF ANSWER

Question 13

21 marks

2023 May/June Paper 42 Question 4

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_s23_ms_42.pdf

4(a)	• proton acceptor
4(b)	• alkali
4(c)	• blue
4(d)	• sodium chloride • water • ammonia
4(e)(i)	• amphoteric (oxides)
4(e)(ii)	• aluminium oxide • or • zinc oxide
4(f)(i)	• all single bonding dot and cross pairs correct • double C = O bond dot and cross pairs are correct • complete diagram is correct
4(f)(ii)	• 3<= pH < 7
4(f)(iii)	• i • CH_3COOH ■ CH_3COO^- + H^+ • H^+ • CH_3COO^- • use of ■
4(f)(iv)	• H^+ + OH^- ■ H_2O
4(g)	• mol KOH = 0.0800 -> 25 / 1000 • = 0.002(00) / 2(00) -> 10-3 • mol H_2SO_4 = / 2 = 0.002 / 2 • = 0.001(00) / 1(00 -> 10-3 • = -> 1000 / 20 = 0.001 -> 1000 / 20 • = 0.05(00) / 5.(00) -> 10-2 • = 98 • = 98 -> = 98 -> 0.05(00) = 4.9(0) (g / dm3)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 14	14 marks
2023 Oct/Nov Paper 42 Question 4	

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 4: Electrochemistry, Topic 12: Experimental techniques and chemical analysis, Topic 2: Atoms, elements and compounds
Answer source label: 0620_w23_ms_42.pdf

4(a)(i)	• lead nitrate (1) • soluble chloride e.g. sodium chloride (1)
4(a)(ii)	• Pb2+(aq) + 2Cl-(aq) -> PbCl2(s) • PbCl2 as only product (1) • Pb2+ + 2Cl- as only reactants (1) • state symbols (1)
4(a)(iii)	• filter (1) • wash (the residue) using water (1) • dry the residue between filter papers / in a warm place (1)
4(b)(i)	• mobile ions
4(b)(ii)	• 2Cl- -> Cl2 + 2e- • any negative Cl species losing electron(s) (1) • correct ionic half equation (1)
4(b)(iii)	• (damp) litmus (paper) (1) • is bleached / goes white (1) • 4biv • (shiny) grey AND solid

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 15

6 marks

2023 Oct/Nov Paper 43 Question 1

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 11: Organic chemistry, Topic 9: Metals, Topic 12: Experimental techniques and chemical analysis, Topic 10: Chemistry of
Answer source label: 0620_w23_ms_43.pdf

1(a)	• potassium iodide
1(b)	• sodium bromide
1(c)	• sulfur dioxide
1(d)	• propene
1(e)	• carbon monoxide
1(f)	• anhydrous copper(II) sulfate

Easy explanation / reminder
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
• Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO2 and H2O when oxygen is in excess.

END OF ANSWER

Question 16

15 marks

2023 Oct/Nov Paper 43 Question 4

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_w23_ms_43.pdf

4(a)(i)	• dilute sulfuric acid
4(a)(ii)	• solid remains / solid undissolved (1) • bubbling / effervescence / fizzing stops (1)
4(a)(iii)	• aqueous zinc sulfate
4(a)(iv)	• any two • zinc oxide • zinc carbonate • zinc hydroxide
4(b)(i)	• (a solution containing the) maximum amount of solute dissolved / no more solute can dissolve (1) • at a given temperature (1)
4(b)(ii)	• solubility decreases as temperature decreases
4(c)(i)	• hydrated
4(c)(ii)	• to ensure all the water of crystallisation is removed
4(c)(iii)	• 2 -> 10-3 / 0.002 (1) • 0.144 (g) (1) • (0.144 ■ 18 =) 8 -> 10-3 / 0.008 (1) • (0.008 ■ 0.002 =) 4 (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 17

16 marks

2024 Feb/March Paper 42 Question 3

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_m24_ms_42.pdf

3(a)	• proton acceptor
3(b)	• a soluble base
3(c)	• blue • colourless
3(d)(i)	• HNO3 • lowest pH
3(d)(ii)	• universal indicator
3(e)	• (CH3COOH) ■ CH3COO- + H+ • H+ • CH3COO- • ■ • February/March 2024
3(f)	• H+ + OH- -> H2O
3(g)	• (0.0150 -> 20.0/1000 =) 0.0003(00) / 3.00 -> 10-4 (mol) • (-> 2 = 3.00 -> 10-4 -> 2 =) 0.0006(00) / 6.00 -> 10-4 (mol) • (-> 1000/25.0 = 6.00 -> 10-4 -> 1000 / 25.0 =) 0.0240 (mol / dm3) • 63 (g / mol) • (-> = 0.0240 -> 63 =) 1.51(2) (g / dm3)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 18

17 marks

2024 May/June Paper 43 Question 5

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_s24_ms_43.pdf

5(a)(i)	• named soluble barium salt e.g. barium chloride / barium nitrate • named soluble sulfate salt • e.g. sodium sulfate / potassium sulfate / ammonium sulfate / sulfuric acid
5(a)(ii)	• filter • wash (residue) with water • and • dry (residue) between filter papers / in a warm place
5(a)(iii)	• Ba2+ (aq) + SO42- (aq) -> BaSO4(s) • BaSO4 as only product • Ba2+ + SO42- as only reactants • state symbols
5(b)	• sulfuric (acid) • copper(II) carbonate / copper(II) oxide / copper(II) hydroxide
5(c)(i)	• glowing splint and relights
5(c)(ii)	• 2CuO + 4NO2 • 6H2O
5(d)(i)	• (To ensure that) all the water is given off
5(d)(ii)	• mol ZnSO4 • (0.322 / 161 =) 0.002(00) • mass H2O given off • (0.574 - 0.322 =) 0.252 • mol H2O given off • (0.252 / 18 =) 0.014(0) • value of x • (0.014 / 0.002 =) 7

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 19

5 marks

2024 Oct/Nov Paper 41 Question 1

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 6: Chemical reactions
Answer source label: 0620_w24_ms_41.pdf

1(a)	• D
1(b)	• F
1(c)	• C
1(d)	• B AND C
1(e)	• B

Easy explanation / reminder
• Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
• escape.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 20

18 marks

2024 Oct/Nov Paper 41 Question 3

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 4: Electrochemistry
Answer source label: 0620_w24_ms_41.pdf

3(a)	• H_2O • state symbols: (s) (aq) (aq)
3(b)	• no more solid dissolves
3(c)	• filtration
3(d)(i)	• substance that is chemically combined with water
3(d)(ii)	• $CuSO_4$-$5H_2O$
3(d)(iii)	• heat • until saturation (point) AND allow to cool
3(e)(i)	• mobile ions
3(e)(ii)	• conducts electricity • inert
3(e)(iii)	• becomes lighter (blue)
3(e)(iv)	• pink AND solid
3(e)(v)	• $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$ • O_2 as a product • OH^- AND e^- • correct equation
3(e)(vi)	• one mark each for any two of: • colour remains constant • no bubbles at the anode • anode dissolves

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 21 — 20 marks

2024 Oct/Nov Paper 43 Question 4

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 3: Stoichiometry, Topic 12: Experimental techniques and chemical analysis
Answer source label: 0620_w24_ms_43.pdf

4(a)(i)	• blue (1) • to • colourless (1)
4(a)(ii)	• too many colour changes
4(b)	• 0.0025 / 2.5 -> 10-3 (mol) • 0.00125 / 1.25 -> 10-3 (mol) • 0.0625 / 6.25 -> 10-2 (mol / dm3)
4(c)	• heat the solution / warm the solution /boil the solution / leave solution in hot place • to saturation (point)/ crystallisation point AND leave to cool • suitable method of drying
4(d)	• KOH + H2SO4 -> KHSO4 + H2O • 4e(i) • acid / acidic
4(e)(ii)	• flame test is carried out on X: • lilac flame • solid copper(II) carbonate is added to X: • solid dissolves / solid disappears • bubbles / fizzing / effervescence • blue solution • aqueous barium nitrate acidified with dilute nitric acid is added to X: • white precipitate
4(f)(i)	• moles of Zn = 0.005 / 5 -> 10-3 (mol) • Zn is limiting because moles of H2SO4 ■ moles of Zn AND 1:1 ratio required for reaction
4(f)(ii)	• (48.0 / 24 000 =) 0.00200 (mol) • 1.20 -> 1021 (molecules)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 22

15 marks

2025 Oct/Nov Paper 42 Question 4

Main topic: Topic 7: Acids, bases and salts
Answer source label: 0620_w25_ms_42.pdf

4(a)(i)	• H+
4(a)(ii)	• complete (dissociation) • partial (dissociation)
4(a)(iii)	• ethanoic (acid)
4(a)(iv)	• colourless
4(a)(v)	• SO42- • CH3COO -
4(a)(vi)	• hydrogen • oxygen
4(a)(vii)	• calcium ethanoate
4(b)(i)	• proton acceptor
4(b)(ii)	• insoluble
4(b)(iii)	• aluminium oxide
4(b)(iv)	• use of -18 or (9 -> -2) or (3 -> 3 -> -2) • +5

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 23

12 marks

2025 Oct/Nov Paper 43 Question 5

Main topic: Topic 7: Acids, bases and salts
Cross-topic links: Topic 12: Experimental techniques and chemical analysis, Topic 9: Metals, Topic 8: The Periodic Table
Answer source label: 0620_w25_ms_43.pdf

5(a)(i)	• Pb2+(aq) + SO42- (aq) -> PbSO4(s) • PbSO4 as the ONLY formula on the right • ONLY Pb2+ and SO42- on the left • state symbols
5(a)(ii)	• lead(II) nitrate
5(a)(iii)	• sodium sulfate / potassium sulfate / ammonium sulfate
5(a)(iv)	• residue
5(a)(v)	• wash with distilled water • suitable method of drying
5(b)(i)	• any two from: • good conductor of electricity • soluble salts • basic oxides
5(b)(ii)	• any two from: • variable oxidation states • coloured compounds • catalyst

Easy explanation / reminder
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER